

# Phase 4 Order Report for LinkedIn post creation:

## Phase 4 — What We Actually Completed

Phase 4 = **Order Creation + Inventory Reservation + Validation + Security + Integration**

Below is the exact breakdown of what was tested and confirmed.

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### 1. Order Creation Flow (Core Business Logic)

#### Verified

- Order creation works end-to-end
- Auto-generated order code

#### Confirmed behavior:

- ORD-YYYYMMDD-XXXX format
  - RAW\_MATERIAL\_ISSUED on creation
  - CONFIRMED stock status
  - Low stock warnings returned correctly
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### 2. Inventory Reservation System

#### Verified

When order is created:

- Stock is calculated from BOM
- Stock is reserved immediately
- Ledger entries are created

#### Confirmed tables:

##### stock\_ledger

- STOCK\_IN entries (initial stock)
- RESERVED entries (negative values)

##### stock\_reservations

- ACTIVE reservation per material
- Correct quantities per BOM

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### 3. BOM Integration (Phase 2 Dependency)

#### Verified

- Multi-material BOM works correctly
  - Each material is processed individually
  - Missing BOM → error (404)
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### 4. Transaction Safety (MOST IMPORTANT)

#### Verified

#### Full rollback scenarios:

- Insufficient stock → rollback everything
- Mid-loop failure → no partial reservation
- BOM missing → no order created

#### Result:

- System is fully **atomic**
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### 5. Validation Layer (DTO Level)

#### Verified Bean Validation:

- customerName required
- cupType required
- cupQuantity:
  - must be  $\geq 1$
- expectedDelivery:
  - required
  - must be future date only

#### Result:

- No invalid request reaches service layer
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## 6. Security (JWT + Role Access)

**Verified:**

### Authentication

- No token → 401

### Authorization

- HR Manager → 403
- Floor Supervisor → 403
- Manager → 201
- Admin → 201

**Result:**

- Role-based access control fully working
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## 7. Inventory Accounting Logic

**Verified:**

- Stock formula works:

`SUM(STOCK_IN + RESERVED)`

- Stock decreases correctly after orders
  - Multiple orders reduce stock cumulatively
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## 8. Edge Case Testing

**Verified:**

- Exact stock exhaustion behavior
  - Sequential orders reduce inventory correctly
  - Negative stock prevented
  - No CONSUMED entries during order creation
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## 9. Integration Coverage

**Covered implicitly:**

- BOM → Order → Inventory linkage
  - Inventory → Ledger → Reservation sync
  - Order → Security → Validation flow
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## **Final Outcome of Phase 4**

### **What Phase 4 proves**

Your system now guarantees:

#### **1. Correct business flow**

Order → BOM → Reservation → Ledger

#### **2. Safe transactions**

No partial state ever persists

#### **3. Correct inventory tracking**

Accurate stock deduction

#### **4. Secure API layer**

Proper role-based access

#### **5. Strict validation**

No invalid data enters system

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### **In one line**

**Phase 4 = Fully working Order + Inventory reservation engine with security + validation + atomic consistency.**

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